



Potato Yield Prediction Research Based on Improved Artificial Neural Networks Using Whale Optimization Algorithm

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Abstract

Potato, as a crucial global staple crop, plays a pivotal role in ensuring global food security. In China, both the cultivation area and yield of potatoes rank among the highest globally, highlighting its significance in agricultural production. Accurate prediction of potato yield is essential for guiding cultivation management and making related decisions. The objective of this study is to develop an improved potato yield prediction model to help address supply gaps in the fresh potato market domestically, particularly between northern and southern regions. Considering the limitations of traditional Backpropagation (BP) neural networks in terms of prediction accuracy and robustness, this study optimizes the BP neural network using the Whale Optimization Algorithm (WOA). By analysing meteorological factors, field hydrothermal factors, and potato yield data collected from field Internet of Things (IoT) systems between 2010 and 2022, we constructed and compared three different models: a traditional BP neural network model, a BP neural network model optimized by Genetic Algorithm (GA), and a BP neural network model optimized by WOA. The research findings indicate that the WOA-BP model significantly outperforms the other two models in prediction accuracy, with an R^2 (coefficient of determination) value of 0.9764. Moreover, the high degree of fit between predicted and observed values validates the scientific validity and accuracy of the WOA-BP model in potato yield prediction.

Keywords Backpropagation neural network · Cultivation management · Potato yield prediction · Whale Optimization Algorithm

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Introduction

Potato, as a significant food and economic crop in China, plays a vital role in ensuring national food security (Wang and Zhang 2004, 2010). Despite the extensive cultivation of potatoes in China, there still exists a gap in yield compared to developed countries, leading to increasingly prominent supply–demand contradictions. Therefore, accurate prediction of potato yield holds significant importance in guiding potato production, stabilizing markets, adjusting agricultural structures, and optimizing field cultivation management in China. Potato yield is influenced by various factors, including field environmental conditions, atmospheric temperature, humidity, light intensity, and rainfall (Haverkort and Kooman 1997; Jannat et al. 2021; Ahoudou et al. 2023; Solihin et al. 2024). The spatial variability of these factors results in yield uncertainty, consequently affecting regional economic development.

Currently, there are various methods for crop yield prediction, including remote sensing techniques (Maddala et al. 2022; Muruganantham et al. 2022), statistical dynamics (Hansen and Indeje 2004; Bustos-Korts et al. 2019), meteorological yield methods (Zhang et al. 2022; Ha et al. 2023), agronomic crop models (Ansarifar et al. 2021; Zhou et al. 2023), mathematical statistics (Marquez-Sanchez 2007; Tian et al. 2020b), and deep learning (Elavarasan and Vincent 2020). Among these, Backpropagation (BP) neural networks, a type of widely applied artificial neural network (ANN), have been utilized by scholars for predicting the yield of various food crops (Tian et al. 2020a; Guo et al. 2022). However, existing studies often overlook the micro-level prediction approach that combines meteorological data with field environmental data, and tend to oversimplify the consideration of influencing factors. Additionally, traditional BP neural networks are prone to problems such as overfitting, local optimum trapping, and slow convergence during the prediction process.

To address these issues, this study proposes a novel approach, utilizing optimization techniques of intelligent optimization algorithms to adjust the initial weights and thresholds of the network, thereby enhancing the model's learning efficiency and global optimization capability to improve network structure. This study selects field environmental data and meteorological data from the Sichuan region, along with potato yield information as experimental data spanning from 2010 to 2022. Through establishing a Whale Optimization Algorithm-Backpropagation (WOA-BP) neural network prediction model, we analyse and compare the accuracy and errors of the traditional BP model and Genetic Algorithm-Backpropagation (GA-BP) model in the training and prediction processes, aiming to provide scientific guidance for the prediction and management of potato production in the Sichuan region.

Materials and Methods

Data Source

This study focuses on potato yield and selects key factors such as atmospheric temperature, atmospheric humidity, rainfall, soil moisture content, and soil temperature as input variables influencing potato yield. Specifically, we choose nine indicators as input features of the network model: monthly maximum soil temperature (X1), monthly minimum soil temperature (X2), monthly average soil temperature (X3), monthly average soil moisture content (X4), monthly maximum atmospheric temperature (X5), monthly minimum atmospheric temperature (X6), monthly average atmospheric temperature (X7), monthly rainfall (X8), and monthly average atmospheric humidity (X9), with potato yield as the output target variable.

To train and validate the model, we use potato yield data from five experimental sites in the Sichuan region between 2010 and 2022 as training samples. Additionally, we select data from one experimental site during the same period as the validation dataset. In the model establishment process, we employ three different models for prediction analysis: BP neural network model, Genetic Algorithm–optimized BP neural network model (GA-BP), and Whale Optimization Algorithm–optimized BP neural network model (WOA-BP). The objective of this study is to compare the prediction results of these three models to identify the most suitable model for predicting potato yield, providing a scientific basis for the prediction and management of potato production in the Sichuan region.

Model Construction

BP Neural Network

In this study, we have employed a four-layer BP neural network topology for model development. The training process of the BP network is primarily divided into three critical steps (Wang et al. 2007; Yang et al. 2021). Initially, meteorological factors and field data are utilized as the input layers. Subsequently, during the forward propagation of the network, errors are calculated using a loss function by comparing the network's output data with the actual measured data. Finally, during the backpropagation of errors, each weight coefficient within the network is updated. After multiple iterations of these three steps and learning the appropriate weights, the model is capable of being applied to new datasets to simultaneously predict multiple target values. Assuming existing sample data for (x, y) , $x = [x_1, x_2, \dots, x_m]$, $y = [y_1, y_2, \dots, y_t]$, the hidden layer neurons are represented as $o = [o_1, o_2, \dots, o_m]$. The network weight matrix between input layer and hidden layer neurons is given by Eq. (1), and the network weights between input layer and output layer neurons are denoted as w^1 and w^3 , respectively:

$$\begin{aligned}
 w^1 &= \begin{bmatrix} w_{11}^1 & w_{12}^1 & \cdots & w_{1d}^1 \\ w_{21}^1 & w_{22}^1 & \cdots & w_{2d}^1 \\ \vdots & \vdots & \cdots & \vdots \\ w_{m1}^1 & w_{m2}^1 & \cdots & w_{md}^1 \end{bmatrix} \\
 w^3 &= \begin{bmatrix} w_{11}^3 & w_{12}^3 & \cdots & w_{1t}^3 \\ w_{21}^3 & w_{22}^3 & \cdots & w_{2t}^3 \\ \vdots & \vdots & \cdots & \vdots \\ w_{b1}^3 & w_{b2}^3 & \cdots & w_{bt}^3 \end{bmatrix}
 \end{aligned} \quad (1)$$

where x_i represents the i -th input value in the neural network and y is the vector of predicted values. a_i^1 represents the i -th activation of the i -th layer neuron, and w_{lm}^i represents the connection between the m -th unit of the i -th layer and the 1st unit of the $(i+1)$ th layer.

The thresholds of hidden layer neurons and output layer neurons are denoted as θ^1 and θ^2 , respectively:

$$\begin{aligned}
 \theta^1 &= [\theta_1^1, \theta_2^1, \dots, \theta_l^1] \\
 \theta^2 &= [\theta_1^2, \theta_2^2, \dots, \theta_n^2]
 \end{aligned} \quad (2)$$

Therefore, the output of hidden layer neurons is:

$$\begin{aligned}
 o_j &= f \left(\sum_{i=1}^m w_{ji}^1 x_i - \theta_j^1 \right) = f(\text{net}_j) \\
 \text{net}_j &= \sum_{i=1}^m w_{ji}^1 - \theta_j^1 \\
 j &= 1, 2, \dots, l
 \end{aligned} \quad (3)$$

And the output of output layer neurons is:

$$\begin{aligned}
 Z_j &= f \left(\sum_{i=1}^l w_{kj}^1 o_i - \theta_k^1 \right) = g(\text{net}_j) \\
 \text{net}_k &= \sum_{i=1}^l w_{kj}^1 o_i - \theta_k^1 \\
 k &= 1, 2, \dots, n
 \end{aligned} \quad (4)$$

where E represents the error between actual output and expected output:

$$E = \frac{1}{2} \sum_{k=1}^n (y_k - z_k)^2 \quad (5)$$

WOA-BP Neural Network

The Whale Optimization Algorithm (WOA) is an optimization technique that simulates the predatory behaviour of whales (Mirjalili and Lewis 2016; Huang et al. 2023). Its workflow mainly consists of three stages. Firstly, in the encircling prey stage, the algorithm regards the nearest whale to the fish group as a local optimal

solution. Other whale individuals calculate their distances from this optimal solution and gradually converge towards it, forming an encirclement around the fish group. The mathematical model for this stage is represented by Eqs. (6) and (7):

$$R = |C \cdot X^*(t) - X(t)| \quad (6)$$

$$X(t + 1) = X^*(t) - A \cdot R \quad (7)$$

Here, A and C represent coefficient vectors; R represents the distance vector; X^* represents the local optimal solution; and X represents the position vector. The mathematical expressions for A and C are given by:

$$A = 2a \cdot r - a \quad (8)$$

where a represents a linearly decreasing vector from 2 to 0; r represents a random vector, $r \in [0,1]$; and A represents the distance coefficient, whose variation is influenced by a .

Secondly, after narrowing the range of fish group activity, whales exhale bubbles to form a ‘bubble net’ to limit the escape of the fish group, thereby searching for the optimal solution within this area. Finally, based on Eq. (8), the fish group is enclosed. When $|A| < 1$, whale individuals gradually approach the optimal solution; the larger $|A|$, the stronger the activity of whale individuals in a larger spatial range, indicating stronger global search capability; the smaller $|A|$, the more meticulous the search of whale individuals in a smaller space, indicating stronger local search capability.

Model Evaluation Metrics

To quantitatively assess the performance of the optimized neural network prediction model, we adopt multiple evaluation metrics, including the coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). These metrics comprehensively reflect the accuracy and reliability of the model predictions, providing important reference for further optimization and application of the model.

Results

Analysis of Model Prediction Results

In this study, through a series of preliminary experiments, we determined the relevant parameters of the BP neural network using a trial-and-error method. Specifically, we set the convergence error to 0.00065, learning rate to 0.05, and maximum training iterations to 50,000. The training process automatically terminates when the network’s convergence error reaches the preset threshold. Based on the collected data, we constructed three different prediction models: the traditional BP neural network model, the Genetic Algorithm–optimized BP neural network model (GA-BP),

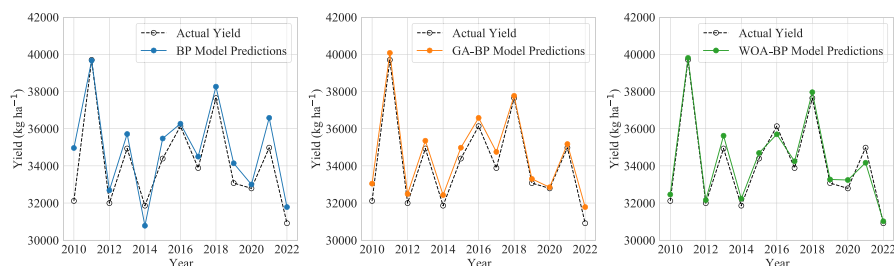


Fig. 1 Prediction curves of neural network models

and the Whale Optimization Algorithm–optimized BP neural network model (WOA-BP). The prediction results of these models for potato yield are shown in Fig. 1.

By comparing the performance of these models in predicting potato yield, we can draw the following conclusions:

- (1) **Prediction accuracy and fitting effect:** From the analysis of Fig. 1, it can be observed that the traditional BP model exhibits significant errors in predicting potato yield. Conversely, the optimized GA-BP and WOA-BP models demonstrate higher prediction accuracy. Particularly, the WOA-BP model exhibits a closer curve fitting between actual and predicted values, indicating superior fitting effect and generalization capability.
- (2) **Convergence and adaptability:** The WOA-BP model not only demonstrates outstanding prediction accuracy but also shows good convergence and adaptability. This implies that the WOA-BP model can maintain stable prediction performance under different datasets and conditions.
- (3) **Computational complexity:** Although the GA-BP model outperforms the BP model in certain aspects of prediction accuracy, its algorithm steps are more complex, resulting in cumbersome computational processes. This may limit the efficiency and feasibility of the GA-BP model in practical applications.
- (4) **Model evaluation metrics:**

Table 1 provides a comprehensive evaluation of the prediction performance for three distinct models: the traditional Backpropagation (BP) neural network model, the Genetic Algorithm–optimized Backpropagation (GA-BP) model, and the Whale Optimization Algorithm–optimized Backpropagation (WOA-BP) model. Neural networks are a type of machine learning (ML) model inspired by biological neural networks, and Backpropagation is a common algorithm used to train these networks.

Table 1 Evaluation of model prediction performance

Model	R^2	RMSE (kg/ha)	MAPE (%)	MAE (kg/ha)
BP	0.8731	1134.4813	2.6703	888.8606
GA-BP	0.9823	602.0105	1.4833	492.1949
WOA-BP	0.9764	422.7631	1.1068	379.8171

Each model's performance is quantified using four key metrics: the coefficient of determination (R^2), the root mean square error (RMSE) measured in kilograms per hectare (kg/ha), the mean absolute percentage error (MAPE) expressed as a percentage, and the mean absolute error (MAE) also measured in kg/ha.

The BP model, which serves as our baseline, achieves an R^2 value of 0.8731, indicating that it can explain 87.31% of the variability in the potato yield data. However, it records an RMSE of 1134.4813 kg/ha and an MAE of 888.8606 kg/ha, suggesting a relatively high level of error in its predictions. The MAPE for the BP model is 2.6703%, reflecting the average magnitude of the percentage error in yield predictions.

The GA-BP model, which utilizes a genetic algorithm to optimize the BP neural network, demonstrates markedly improved performance. It achieves an R^2 value of 0.9823, signifying that it can explain 98.23% of the variability in the data—a substantial improvement over the baseline BP model. This model reduces the RMSE to 602.0105 kg/ha and the MAE to 492.1949 kg/ha, showing a notable decrease in prediction errors compared to the BP model. The MAPE is also reduced to 1.4833%, indicating more accurate predictions overall.

The WOA-BP model, which employs the Whale Optimization Algorithm to enhance the BP neural network, exhibits the best performance among the three. While its R^2 value of 0.9764 is slightly lower than that of the GA-BP model, it still represents a significant improvement over the baseline BP model. Notably, this model further lowers the RMSE to 422.7631 kg/ha and the MAE to 379.8171 kg/ha, indicating the most accurate predictions among all three models. The MAPE is the lowest at 1.1068%, which represents a substantial improvement over both the BP and GA-BP models.

In conclusion, the empirical results demonstrate a clear performance hierarchy among the three models. The WOA-BP model consistently outperforms both the BP and GA-BP models in terms of prediction accuracy and error minimization, as evidenced by the lower RMSE, MAE, and MAPE values. These findings underscore the effectiveness of the Whale Optimization Algorithm in enhancing the predictive capabilities of Backpropagation neural networks for potato yield prediction tasks. The significant improvements observed in the optimized models (GA-BP and WOA-BP) over the baseline BP model highlight the potential of hybrid algorithms that combine evolutionary optimization techniques with traditional neural network architectures to enhance predictive modeling in agricultural contexts.

Analysis of Model Prediction Errors

To gain a deeper understanding of the performance of different models in predicting potato yield, we conducted a comprehensive analysis of the prediction errors of each model. Figure 2 displays the distribution of prediction errors for different models. In this figure, the horizontal axis represents the errors between predicted and actual values, while the vertical axis represents the number of prediction samples in the training set. The error distribution is centred around zero, with deviations spreading on both sides. The zero axis represents the perfect agreement between predicted

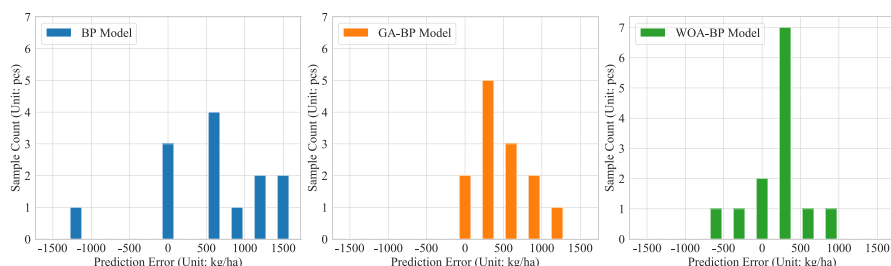


Fig. 2 Distribution of prediction errors for neural network models

and actual values. Closer proximity to the zero axis indicates smaller differences between predicted and actual values, reflecting higher prediction accuracy of the model.

By comparing the error distributions in Fig. 2, it can be observed that the WOA-BP model has more sample points with errors close to zero compared to the BP and GA-BP models, and overall, a smaller range of errors. This indicates that the WOA-BP model exhibits higher accuracy in predicting potato yield. Among the various combination prediction models, the WOA-BP model has the highest number of sample points and smallest errors in the zero-error interval, whereas other models have fewer sample points and relatively larger errors in this interval.

This result further confirms the superiority of the WOA-BP model in prediction accuracy. The integration of the Whale Optimization Algorithm significantly enhances the optimization process of weights and thresholds in the BP neural network, effectively reducing prediction errors and enhancing the model's generalization capability. This optimization not only improves the performance of the model on the training set but also provides higher reliability for the model's application on unknown data.

Conclusion

This study applied the BP neural network to predict potato yield and found certain limitations in the traditional BP model, such as unstable prediction performance and significant errors in some predictions. Specifically, the correlation coefficient (R^2) between predicted and actual values in the BP model was only 0.8731, indicating the need for improvement in model explanatory power and prediction accuracy.

To address this issue, we constructed a Whale Optimization Algorithm (WOA)-optimized BP neural network model (WOA-BP). By optimizing the network's weights and thresholds using the WOA algorithm, we successfully identified the optimal weight combinations. After multiple iterations of training, the prediction accuracy of the WOA-BP model significantly improved, with R^2 reaching 0.9764, mean absolute percentage error (MAPE) reduced to 1.10668%, and root mean square error (RMSE) at 422.7631. This result indicates that the prediction accuracy of the WOA-BP model meets the practical operational requirements of potato production,

which is of significant practical importance for guiding planting management practices and agronomic measures during potato growth.

In summary, the WOA-BP model demonstrates excellent performance in predicting potato yield, not only enhancing prediction accuracy and stability but also providing strong scientific support for agricultural production management and decision-making. Future research can further explore the optimization space of the model and the possibility of applying the model to predict yields of other crops, thereby providing more technical support and decision-making references for sustainable agriculture development.

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Data Availability On reasonable request, the corresponding author will provide data supporting the study’s results. The raw data cannot be made public for reasons of confidentiality and privacy. However, researchers who satisfy the requirements for access to confidential data can be given access to aggregated and anonymized data as well as statistical analysis codes. To request access to the data, interested researchers should contact the corresponding author.

Declarations

Informed Consent Not applicable.

Conflict of Interest The authors declare no competing interests.

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